Term Project Guidelines

EE5203 Microprocessors — 2026-2027 Fall

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1 What You Will Build

In a team of **2 or 3 students**, you will build an embedded system on the **ST Nucleo-F401RE** board. The system reads data from at least one sensor, processes it, shows or uses the result, and communicates over a serial bus. You develop it step by step through milestones that are checked in your lab sessions, and you finish with a demonstration and an individual oral defence. The project is **20%** of your course grade.

2 Key Dates

Every team has a **home section**: the lab section you name when you register (Lab 1: Tuesday 13:00, Lab 2: Tuesday 15:00, Lab 3: Wednesday 13:00). Your milestones are checked in the lab of your home section.

Week	Milestone	What you show	Date
5	Team registration	E-mail your team members (names, student numbers, lab sections), your home section and your topic to the milestone assistant of your home section	**18 October 2026, 23:59**
6	Proposal	One-page PDF on CATS: problem, block diagram, components, minimum working system	Before the lab of your home section, **20/21 October**
9	Design review and first hardware	Design document on CATS before the lab: pin and peripheral table, data flow, who does what, risks. In the lab: one input or output part working	Lab, **10/11 November**
10	Data acquisition	Sensor data read on the board and checked against a known value	Lab, **17/18 November**
11	Communication	UART log or command interface (or your I2C/SPI device) working	Lab, **24/25 November**
12	Submission	Report, slides and source code on CATS	**29 November 2026, 23:59**
12	Demonstration and oral defence	Live demonstration, presentation and individual defence, in one joint session for all teams in our laboratory room	**Tuesday 1 December, from 13:00**

There is no make-up for a missed milestone or the demonstration, and late submissions are not accepted.

Milestone assistants. The milestone assistant of your home section registers your team, receives your uploads and checks your milestones:

- Lab 1 and Lab 2: Res. Asst. Ahmet Aytuğ Ayrancı (ahmet.ayranci@iku.edu.tr)
- Lab 3: Res. Asst. Ahmet Can Tanık (a.tanik@iku.edu.tr)

How the checks in weeks 9, 10 and 11 work. The check takes the first part of the lab of your home section. Right after the opening brief, your milestone assistant calls the teams one at a time, in team-number order, for a 3-minute check: you show the milestone working on your hardware. At least one team member must be there with the hardware; members from other sections attend their own lab as usual. Have your hardware connected and your code flashed before your team is called. A team that is not ready when called misses the milestone. The result is pass or miss. The rest of the lab runs as usual.

3 Rules for Teams and Topics

- A team has **2 or 3 members**. Members may come from different lab sections; the team names one **home section** when it registers.
- Choose a topic from Section 9 or propose your own. Your proposal in week 6 must be approved before you continue.
- No two teams in the course can build the same project. The first team to register a topic gets it.
- After the proposal you cannot change your team or topic without the instructor's approval.
- **Parts:** the course kit has no sensors. Your team buys the sensors, displays and actuators your project needs. Order them **before the proposal**; parts that arrive late do not move a milestone. Until a part arrives, you may use the potentiometer as a stand-in if you say so.

4 Technical Requirements

Your system **must** have all of these parts:

1. **Board and tools:** Nucleo-F401RE, programmed in C with STM32CubeMX and VS Code (HAL or register level).
2. **Data acquisition:** at least one sensor, read through the ADC or a digital interface.
3. **Processing:** a clear method that turns raw data into a decision or a result (for example averaging and thresholds, a state machine, a controller).
4. **Output:** show or use the result on a suitable device (display, LEDs, buzzer, servo or motor).
5. **Communication:** at least one serial protocol — UART, I2C or SPI — between the board and a sensor, a display or a PC.

Minimum complexity. A sensor connected to one LED is not enough. Your system must use **at least three** of the course topics below, and its timing must not depend on `HAL_Delay()` in the main loop:

Course topic	Lecture
External interrupts (EXTI)	L06
Timers and non-blocking timing	L07
PWM, output compare or input capture	L08
ADC	L09
UART	L10
I2C or SPI	L11

5 What You Submit

Upload **one ZIP file per team** to CATS by 29 November, 23:59. Name it `EE5203_Project_TeamNN.zip`. It contains:

1. **Source code:** the CubeMX project (`.ioc` file and the `Core` folder). Do not include the `build` folder.
2. **Report (PDF, 10-15 pages)** using the report template on CATS. It must contain:
   - the **requirements table**: each requirement, how you test it, and the measured result;
   - the **resource map**: every pin, timer, interrupt and peripheral you use, and what it does;
   - the **timing budget**: how often each task runs and how long it takes;
   - design decisions, wiring diagram, evidence (photos, logs, measurements) and known issues;
   - **work distribution**: which member wrote which part;
   - **AI use**: which tools you used and for what (see Section 7). Write "none" if you used none.
3. **Slides (PDF): 6 to 8 slides per team, not counting the title slide.** Every member presents at least 2 of them.

Keep a working copy of your final code that you can flash in under two minutes on the day.

6 Demonstration and Oral Defence

All teams demonstrate in **one joint session on Tuesday 1 December, from 13:00, in our laboratory room**. The running order is announced on CATS. Every member must be present for the whole session.

15 minutes per team in total, presentation, demonstration and questions included. Set up while the team before you presents: your system must already be running when your slot starts. A slot that runs over is stopped.

Time	What you do
0-8 min	Presentation with live demonstration: every member presents at least 2 slides; show each requirement on the running system as you present it (one requirement, one test, one observed result)
8-9 min	**One fault you inject on purpose** (say first what you expect)
9-15 min	Individual questions: about 3 minutes per member in a team of two, 2 minutes in a team of three

Your known issues belong in the report; you may be asked about them.

Individual questions. Each member answers short questions about their own contribution, at up to three levels:

1. **Your contribution:** "Which parts did you write? Walk me through one."
2. **The mechanism:** "Which register does that configure? What does the hardware do with it?"
3. **The alternative:** "Why this way? What would happen if you did it differently?"

Be ready to open the code and the register view and point to the answer.

7 Use of AI Tools

These rules come from the course syllabus.

- **Allowed:** asking an AI tool to explain a concept, a compiler or debugger message, or a datasheet passage; help with report wording; review of code you wrote yourself.
- **Not allowed:** generating project code; producing the analysis or conclusions of the report with AI.
- **Disclosure:** state in the report how you used AI tools.
- **Check:** the oral defence. Work you cannot explain and change yourself earns no credit.

8 Grading

Part	Points	Graded per
Demonstration against your requirements	35	team
Report	25	team
Presentation (the slides you present)	15	member
Individual oral defence	25	member

Milestones. Each milestone in weeks 6, 9, 10 and 11 that is missed costs the team **5 points**. A milestone is missed when the upload is not on CATS when the lab of your home section starts, or when the check in that lab does not pass.

Team points = demonstration + report (at most 60). **Your share of the team points depends on your defence:**

Your project grade = (team points – milestone deductions) × defence multiplier + your presentation points + your defence points

Your result in the oral defence	Defence multiplier
You explain your contribution **and** the hardware mechanism behind it	× 1
You explain your contribution, but not the mechanism (register, peripheral)	× 0.5
You cannot explain your own contribution	× 0 (project grade = 0)

Example. A team scores 30 (demonstration) + 20 (report) = 50 team points and missed no milestone.

Member	Presentation and defence	Calculation	Project grade
A	presentation 12; explains contribution and mechanism, defence 22	50 × 1 + 12 + 22	84
B	presentation 10; explains contribution, not the mechanism, defence 12	50 × 0.5 + 10 + 12	47
C	cannot explain own contribution	0	0

9 Project Ideas

Each idea lists one possible set of parts and course topics, and how you can check your measurement against a known value. You may change the parts or propose your own idea.

#	Project	Sensors and outputs	Course topics	Check against
1	Parking assistant	HC-SR04 ultrasonic sensor; buzzer; LED bar or display	input capture, PWM, timers, UART	tape measure
2	Room climate monitor with alarm	BME280 or SHT31 sensor (I2C); I2C OLED or LCD; buzzer	I2C, timers, EXTI (alarm reset), UART log	room thermometer
3	Line-following robot	IR line sensors; TB6612FNG motor driver; robot chassis	ADC or GPIO, PWM, timers, EXTI (start/stop), UART tuning log	sensor values on black and white
4	Tilt-stabilised camera platform	MPU6050 IMU; two hobby servos (two-axis gimbal)	I2C, PWM, timers (control loop), UART tuning	protractor angles
5	Sunrise alarm clock	DS3231 real-time clock; light sensor (LDR); LED that brightens before the alarm; display; buzzer	I2C, ADC, PWM, EXTI (buttons), timers	phone clock; covered and lit LDR
6	Motion alarm system	PIR sensor; keypad or buttons for the code; buzzer; display	EXTI, timers, PWM (siren), UART event log	walk-past tests at set distances
7	Vibration and tilt monitor	ADXL345 or MPU6050 accelerometer; vibration motor as a test source; buzzer	I2C or SPI, timers (fixed sampling rate), EXTI (data-ready pin), UART stream to a PC plot	1 g when flat, 0 g on its side
8	Digital weighing scale	Load cell with HX711 amplifier; display; tare button	timers, EXTI (tare), I2C display, UART log	known weights (water bottle, coins)
9	Temperature-controlled fan	LM35 or NTC thermistor; DC fan with transistor; LCD	ADC, PWM (fan speed), timers, I2C display	room thermometer
10	Colour sorter	TCS34725 colour sensor; servo that sorts objects into bins	I2C, PWM (servo), timers, UART	sorting reference cards
11	Motor speed controller	DC motor with encoder or slotted speed sensor; motor driver	input capture (speed), PWM, timers (control loop), UART commands (set speed)	phone stroboscope or tachometer app
12	Ultrasonic radar scanner	HC-SR04 on a servo that sweeps 0-180°; PC plot of angle and distance	PWM, input capture, timers, UART to PC	objects at measured positions
13	Pulse-rate monitor	MAX30102 (I2C) or analog optical pulse sensor; OLED display	ADC or I2C, timers (sampling), I2C display, UART log	pulse counted by hand for 30 s
14	Plant watering controller	Capacitive soil-moisture sensor; small pump with transistor; display	ADC, timers, PWM or GPIO (pump), UART commands	dry soil, wet soil, glass of water

Electrical rules. The pins work at 3.3 V: a 5 V sensor output needs a voltage divider or a level shifter. Motors, pumps and fans need a driver with a flyback diode and their own supply, never a pin. No mains (230 V) wiring. The pulse-rate monitor is a demonstration, not a medical device.